COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures & Algorithms	**SECTION:**	CPE21S4
DATE PERFORMED:	Jul 15, 2026	**DATE SUBMITTED:**	Jul 16, 2026
NAME:	Bal, Kevin Ranz J.	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 2.1 [Memory Diagram Exercise]

1. PROCEDURE OUTPUT

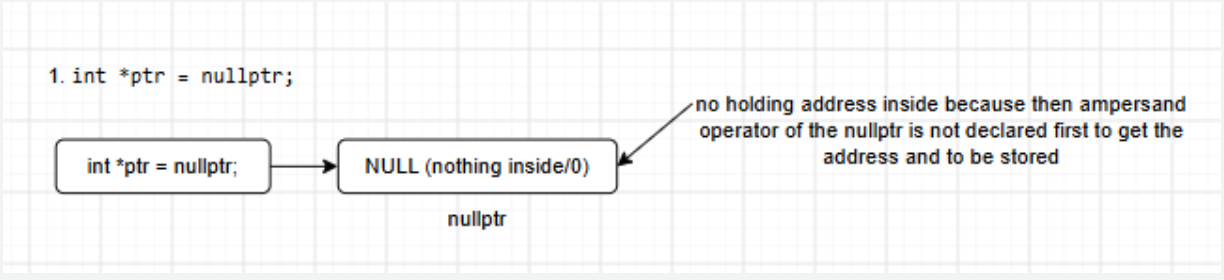
What procedures did you perform?

Performed to create a diagram about pointers, address, heap memory, dynamic allocation, and the question about the constructor and the difference between the three functions.

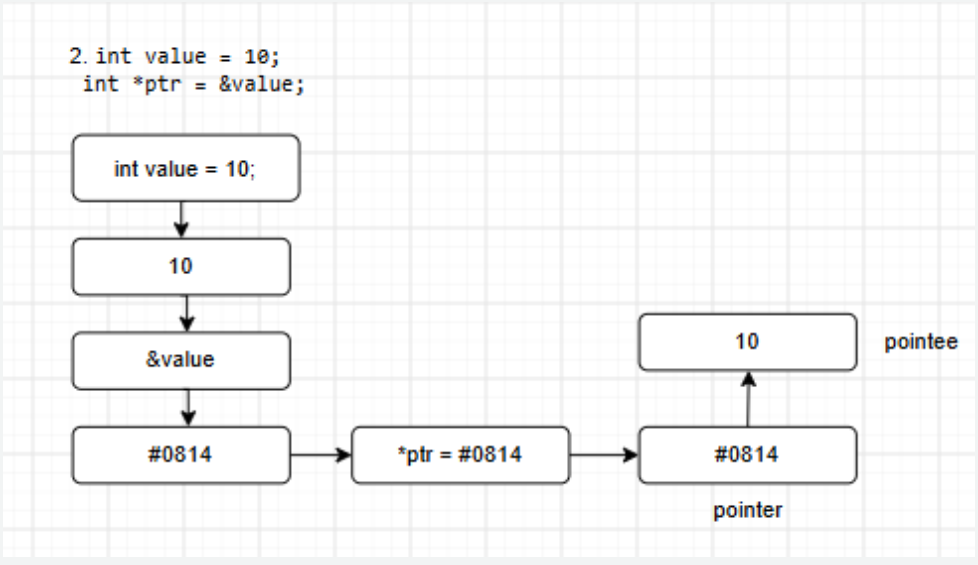
2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

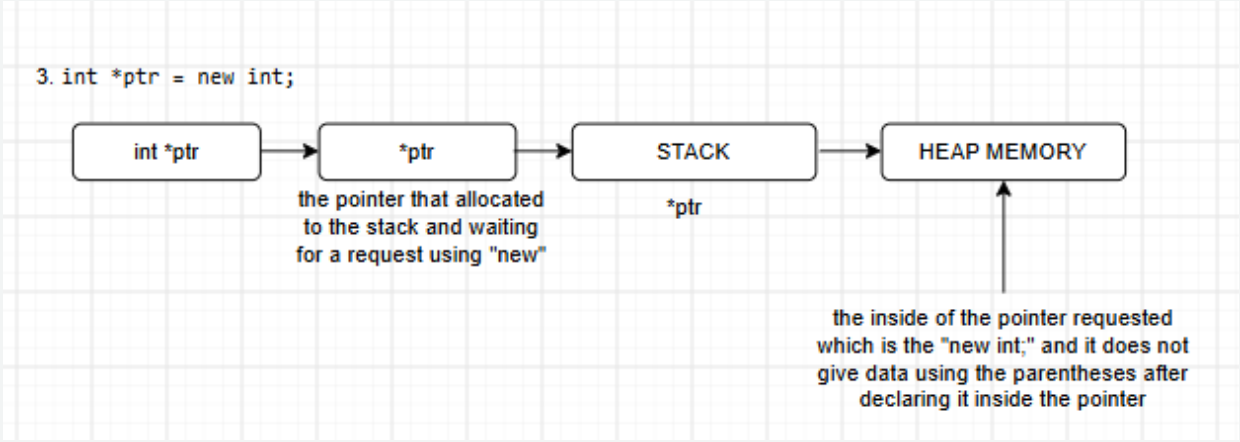
```
1. int *ptr = nullptr;
```



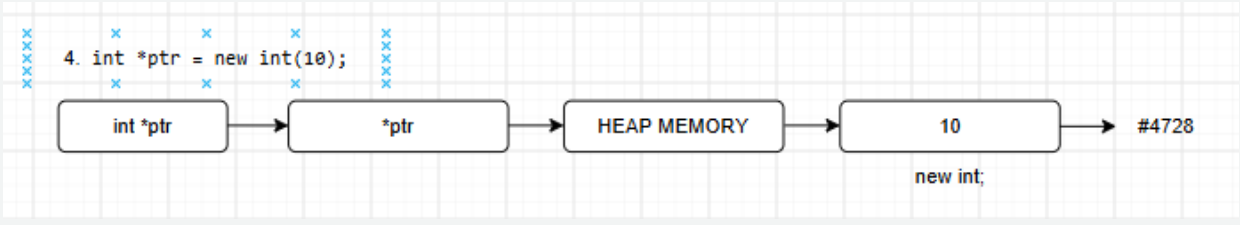
```
2. int value = 10;
   int  *ptr= &value;
```




```
3. int *ptr = new int;
```




```
4. int *ptr = new int(10);
```



5. Differentiate Constructor, Destructor, Copy Constructor, Copy Assignment Operator.

Constructor creates an object and the object depends if wants to add a parameters inside and once called of the object of the class was already created, and Destructor destroys the objects after use and to release resources like dynamic allocated memory while the Copy Constructor creates another object as a copy of the another one which is the already created first.Copy Assignment copies the existing values of the existing object

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.